CST4990 - Dissertation Proposal Form

Date of Submission: 29/05/2026

Name	NOURA LAKRIMDI
Student Id	M00845271
Programme of Study	MSC DATA SCIENCE
Project Title	Geographic Variation in Large Language Model Responses to Mental Health Queries
Campus	HENDON
Supervisor Name	DR. GIOVANNI QUATTRONE
Supervisor Approval	Yes

Supervisor Signature	DR. GIOVANNI QUATTRONE

Section 1: Academic

This section helps academic staff evaluate your project's viability and assign a suitable supervisor. It will also serve as a basis for workshop discussions with your supervisor.

STUDENT NAME: NOURA LAKRIMDI
STUDENT NUMBER: M00845271
PROPOSED TITLE OF PROJECT: Geographic Variation in Large Language Model Responses to Mental Health Queries.
BRIEFLY DISCUSS THE PROBLEM THAT YOU ARE INVESTIGATING AND CITE RELEVANT REFERENCES. (max 300 words) Large language models (LLMs) such as GPT-4o, Claude, and Gemini are increasingly used as informal sources of mental health support and health-related information (Hua et al., 2025; Mesquiti & Nook, 2026). Recent studies show that users frequently interact with conversational AI systems for emotional guidance, psychological advice, and healthcare-related questions (Rousmaniere et al., 2025). At the same time, research has demonstrated that LLM-generated responses can vary in quality, accuracy, and safety across healthcare contexts (Ayers et al., 2023; Wang & Zhang, 2024; Busch et al., 2025). Although existing work has explored demographic disparities and bias in medical large language models (Omar et al., 2024), limited attention has been given to geographic variation in mental health responses. In particular, it remains unclear whether LLMs provide systematically different levels of support depending on the user's implied location, especially across regions with varying healthcare infrastructure and socioeconomic conditions. Cross-lingual and cultural studies have further suggested that LLM performance may differ significantly across contexts and languages (Jin et al., 2024), raising concerns about fairness and consistency in the global deployment of AI-based mental health tools. This project investigates whether large language models generate geographically differentiated mental health guidance and whether responses contain implicit cultural assumptions or unequal levels of actionability. The study will compare responses generated by multiple LLMs using standardised mental health prompts across different cities worldwide. By combining computational NLP analysis with structured human evaluation, the research aims to contribute to ongoing discussions surrounding AI fairness, digital health, and the responsible deployment

of large language models in sensitive healthcare settings.

GIVE YOUR RESEARCH QUESTIONS (Max 3) AND/OR AIMS & OBJECTIVES

Research Question:

1. Do large language models generate systematically different mental health responses across geographic contexts?
2. Do large language models provide less actionable or less specific mental health support in lower-resource settings compared with higher-resource settings?
3. What implicit cultural assumptions appear in geographically contextualised mental health responses generated by large language models?

Aim

This study aims to investigate whether large language models generate geographically differentiated mental health responses and to evaluate whether variations in response quality, specificity, and cultural assumptions emerge across different global contexts.

Objectives

1. To collect and generate a standardised dataset of mental health responses from GPT-4o, Claude, and Gemini using prompts associated with different geographic locations.
2. To evaluate differences in response quality, specificity, actionability, and crisis referral guidance across geographic contexts.
3. To identify and analyse implicit cultural assumptions and patterns present in LLM-generated mental health responses.
4. To compare model responses using both automatic NLP-based analysis and structured human evaluation methods.
5. To examine whether variations in healthcare infrastructure and socioeconomic context are associated with differences in LLM-generated mental health support.

IDENTIFY YOUR RESEARCH METHODS

This study will adopt a mixed-methods research approach combining both quantitative and qualitative analysis. Data will be collected by generating responses from multiple large language models (GPT-4o, Claude, and Gemini) using a standardised set of mental health prompts associated with different geographic locations.

Quantitative methods will include automated NLP-based analyses such as response length analysis, named entity recognition, semantic similarity analysis, and comparisons of referral patterns across locations. Statistical analysis will be used to identify systematic differences between models and geographic contexts.

Qualitative methods will include structured thematic coding and human evaluation of responses to assess actionability, crisis referral quality, specificity, and implicit cultural assumptions. Inter-rater reliability testing using Cohen's kappa will be conducted to ensure consistency in qualitative coding.

The project also incorporates elements of AI auditing and comparative evaluation by systematically assessing how different large language models respond to equivalent mental health queries across diverse global settings.

Data Collection

Data will be collected by constructing a standardised set of mental health prompts representing common emotional distress scenarios such as hopelessness, anxiety, stress, and isolation. These prompts will be submitted to multiple large language models, including GPT-4o, Claude, and Gemini.

The geographic context of the user will be systematically varied by associating each prompt with different cities across multiple countries, representing diverse socioeconomic conditions and healthcare infrastructures. Responses generated by the models will be collected through the respective LLM APIs and stored in a structured dataset for analysis.

Additional contextual data on healthcare infrastructure and mental health service availability will be obtained from publicly available sources, such as the WHO Mental Health Atlas. The final dataset will include prompt text, geographic context, model type, generated responses, and extracted evaluation features for subsequent quantitative and qualitative analysis.

Data Structure and Alignment

The dataset will be organised in a structured tabular format, with each row representing a single LLM-generated response. Key variables will include the prompt identifier, mental health scenario category, geographic location, country income level, model type (GPT-4o, Claude, or Gemini), generated response text, and extracted evaluation metrics. Additional external indicators, such as healthcare infrastructure and mental health service availability, will be aligned using publicly available datasets, including the WHO Mental Health Atlas. Responses will be standardised and cleaned to ensure consistency across models and locations before analysis.

Quantitative Data Analysis:

Quantitative analysis will be conducted using NLP-based metrics and statistical comparisons across models and geographic contexts. Measures such as response length, referral frequency, named entity occurrence, and semantic similarity will be extracted from the generated responses. Descriptive statistics and visualisations will be used to identify trends and differences across locations and models. Statistical tests will be applied to examine whether observed differences are significant across healthcare infrastructure levels and socioeconomic categories.

Statistical Approach:

The study will use descriptive statistics, comparative analysis, and inferential statistical techniques to evaluate differences in LLM-generated responses. Statistical methods may include correlation analysis, ANOVA, the Kruskal-Wallis test, and regression analysis, depending on the data distribution and variable types. Inter-rater reliability for qualitative coding will be assessed using Cohen's kappa coefficient to ensure consistency between human evaluators.

Machine Learning Approach:

The project will apply natural language processing techniques to analyse and compare responses from large language models across geographic contexts. Methods may include named entity recognition, semantic embedding analysis, sentiment analysis, and thematic clustering. Pre-trained transformer-based NLP models and libraries, such as spaCy, sentence-transformers, and scikit-learn, will be used to support automated text analysis and feature extraction.

Comparative Evaluation:

The study will compare responses generated by GPT-4o, Claude, and Gemini using identical mental health prompts across multiple geographic settings. Evaluation will focus on response specificity, actionability, crisis referral quality, and implicit cultural assumptions. Comparative analysis will be conducted across models, cities, and healthcare infrastructure categories to identify systematic differences in response behaviour and potential geographic disparities.

WHAT ARE THE PROBABLE PROJECT OUTCOMES?

Probable Project Outcomes:

The project is expected to produce a systematic evaluation framework for analysing geographic variation in responses from large language models to mental health queries. The study will likely identify whether significant differences exist in response quality, specificity, crisis referral guidance, and cultural assumptions across different geographic and socioeconomic contexts.

The research is also expected to generate a structured dataset of LLM-generated mental health responses across multiple cities and models, which may support future research in AI fairness and digital health. In addition, the project may provide methodological recommendations for evaluating large language models in sensitive healthcare settings using mixed-methods analysis.

The findings could contribute practical recommendations for the safer and more equitable deployment of AI-based mental health support systems, particularly in regions with limited healthcare infrastructure. The study may also highlight potential risks associated with implicit cultural stereotyping and unequal access to actionable guidance in AI-generated responses.

IDENTIFY ANY TECHNICAL RESOURCES NEEDED *(e.g., hardware, software)*

Technical Resources Needed:

The project will require access to a computer or laptop capable of running Python-based data analysis and NLP workflows. Cloud-based large language model APIs, including GPT-4o, Claude, and Gemini, will be required for generating responses to mental health prompts.

Software resources will include Python, Jupyter Notebook, and Overleaf for coding, analysis, and documentation. Key Python libraries are expected to include pandas, NumPy, scikit-learn, spaCy, transformers, sentence-transformers, matplotlib, and seaborn.

The project will also require internet access for API communication and access to publicly available datasets such as the WHO Mental Health Atlas. Version control and data storage tools such as GitHub and Google Drive may also be used for project organisation and backup.

IDENTIFY ANY POSSIBLE CONSTRAINTS?

Possible Constraints:

One potential constraint is the cost and access limitations associated with commercial large language model APIs such as GPT-4o, Claude, and Gemini. API rate limits or pricing restrictions may affect the number of responses that can be generated during the study.

Another challenge is the variability and non-determinism of LLM outputs, as models may generate slightly different responses to the same prompt. This may require repeated sampling and careful standardisation during data collection and analysis.

The study may also face limitations related to the availability and accuracy of geographic mental health infrastructure data across different countries. Public datasets may contain incomplete or inconsistent information depending on the region.

Qualitative evaluation and thematic coding may introduce subjectivity during the interpretation of responses. To reduce this risk, structured coding frameworks and inter-rater reliability testing will be applied.

Additional constraints may include computational requirements for NLP analysis, time limitations within the project schedule, and ethical considerations related to analysing sensitive mental health content and potential cultural stereotyping in AI-generated responses.

GIVE A PROJECT SCHEDULE *(Proposed progress by each week until submission date.)*

• Weeks 1–2: Literature review and data familiarisation

• Weeks 3–4: Data preprocessing and exploratory analysis • Weeks 5–7: Method development and modelling • Weeks 8–9: Evaluation and visualisation • Week 10: Final write-up and presentation preparation
(staff use only) Project Approved by Academic Team? Please tick YES ☐ NO ☐
Academic Staff comments:

Section 2: Ethics Review and Approval

This Research Ethics Screening Form will enable taught students to self-assess and determine whether their research requires ethical review and approval via the online Ethics Monitor before commencing their study. Supervisors must approve this form after consultation with students.

Please answer whether your research/study involves any of the following:		
1. [H]ANIMALS or animal parts.	☐ Yes	X No
2. [M]CELL LINES (established and commercially available cells - biological research).	☐ Yes	X No
3. [H]CELL CULTURE (Primary: from animal/human cells- biological research).	☐ Yes	X No
4. [H]CLINICAL Audits or Assessments (e.g. in medical settings).	☐ Yes	X No
5. [X]CONFLICT of INTEREST or lack of IMPARTIALITY. If unsure, see "Code of Practice for Research" (Sec 3.5) at: https://unihub.mdx.ac.uk/study/spotlights/types/research-at-middlesex/research-ethics	☐ Yes	X No
6. [X]DATA to be used that is not freely available (e.g. secondary data needing permission for access or use).	☐ Yes	X No
7. [X]DAMAGE (e.g., to precious artefacts or to the environment) or present a significant risk to society).	☐ Yes	X No
8. [X]EXTERNAL ORGANISATION – research carried out within an external organisation, or your research is commissioned by a government (or government body).	☐ Yes	X No
9. [M]FIELDWORK (e.g., biological research, ethnography studies).	☐ Yes	X No
10. [H]GENETICALLY MODIFIED ORGANISMS (GMOs) (biological research).	☐ Yes	X No

11. [H]GENE THERAPY including DNA sequenced data (biological research).	☐ Yes	X No
12. [M]HUMAN PARTICIPANTS – ANONYMOUS Questionnaires (participants not identified or identifiable).	☐ Yes	X No
13. [X]HUMAN PARTICIPANTS – IDENTIFIABLE (participants are identified or can be identified): survey questionnaire/ INTERVIEWS/focus groups/experiments/observation studies/ evaluation studies.	☐ Yes	X No
14. HUMAN TISSUE (e.g., human relevant material, e.g., blood, saliva, urine, breast milk, faecal material).	☐ Yes	X No
15. [ILLEGAL]/HARMFUL activities research (e.g., development of technology intended to be used in an illegal/harmful context or to breach security systems, searching the internet for information on highly sensitive topics such as child and extreme pornography, terrorism, use of the DARK WEB, research harmful to national security).	☐ Yes	X No
16. [X]PERMISSION is required to access premises or research participants.	☐ Yes	X No
17. [X]PERSONAL DATA PROCESSING (Any activity with data that can directly or indirectly identify a living person). For example, data gathered from interviews, databases, digital devices such as mobile phones, social media or internet platforms or apps, with or without individuals'/owners' knowledge or consent, and/or could lead to individuals/owners being IDENTIFIED or SPECIAL CATEGORY DATA (GDPR[1]) or CRIMINAL OFFENCE DATA. [1]Special category data (GDPR- Art. 9):" personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership, and the processing of genetic data, biometric data for uniquely identifying a natural person, data concerning health or data concerning a natural person's sex life or sexual orientation".	☐ Yes	X No
18. [X]PUBLIC WORKS DOCTORATES: Evidence of permission is required for use of works/artefacts (that are protected by Intellectual Property (IP) Rights, e.g. copyright, design right) in a doctoral critical commentary when the IP in the work/artefacts was jointly prepared/produced or is owned by another body	☐ Yes	X No
19. [H]RISK OF PHYSICAL OR PSYCHOLOGICAL HARM (e.g., TRAVEL to dangerous places in your own country or in a foreign country (see https://www.gov.uk/foreign-travel-advice), research with NGOs/humanitarian groups in conflict/dangerous zones, development of technology/agent/chemical that may be harmful to others, any other foreseeable dangerous risks).	☐ Yes	X No
20. [X]SECURITY CLEARANCE – required for research.	☐ Yes	X No
21. [X]SENSITIVE TOPICS (e.g., anything deeply personal and distressing, taboo, intrusive, stigmatising, sexual in nature, potentially dangerous, etc).	☐ Yes	X No

M – Minimal Risk; X – More than Minimal Risk. H – High Risk

If you have answered 'Yes' to ANY of the items in the table, your application REQUIRES ethical review and approval using the online **Ethics Monitor BEFORE commencing your research**. Please submit the ethical review and approval request through the **Ethics Monitor**. Further information, such as links to **Ethics Monitor** and training videos, is available at MyLearning(*Middlesex Online Research Ethics) area*: https://mdx.mrooms.net/course/view.php?id=12277 (Login required).

If you have answered 'No' to ALL items in the table, your application is Low Risk, and you may NOT require ethical review and approval via **Ethics Monitor** before commencing your research. Your research supervisor will confirm this below.

Student Signature noura lakrimdi Date:29-05-2026

To be completed by the supervisor:

Based on the details provided in the self-assessment form, I confirm that:	Insert Y or N
The study is Low Risk and *does not require* ethical review and approval using **Ethics Monitor.**	Y
The study *requires* ethical review and approval using **Ethics Monitor**.	N

Supervisor Signature: Date: